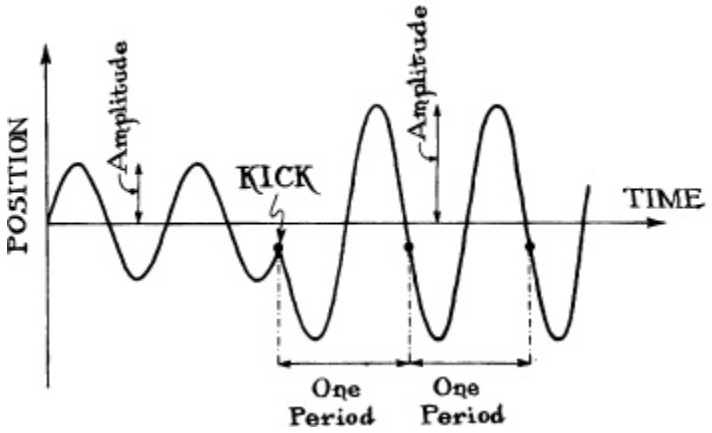
will change, but no matter what that amplitude change is, precisely one period of oscillation after the kick the bar's vibrating end will return to the same position as it had at the moment of the kick (black dots in Figure 10.5). At least this is true if a gravitational wave (or some other force) has not squeezed or stretched the bar in the meantime. If a wave (or other force) *has* squeezed the bar in the meantime, then the bar's position one period later will be changed.

To detect the wave, then, one should build a sensor that makes stroboscopic measurements of the bar's vibrating ends, a sensor that measures the position of the bar's ends quickly once each period of vibration. Such a sensor will kick the bar in each measurement, but the kicks will not change the position of the bar's ends at the times of subsequent measurements. If the position is found to have changed, then a gravitational wave (or some other force) must have squeezed the bar.

Although quantum nondemolition sensors solved the problem of Braginsky's standard quantum limit, by the mid-1980s I had become rather pessimistic about the prospects for bar detectors to bring gravitational-wave astronomy to fruition. My pessimism had two causes.



10.5 The principle underlying a stroboscopic quantum nondemolition measurement. Plotted vertically is the position of the end of a vibrating bar; plotted horizontally is time. If a quick, highly precise measurement of the position is made at the time marked KICK, the sensor that makes the measurement will give the bar a sudden, unknowable kick, thereby changing the bar's amplitude of vibration in an unknown way. However, there will be no change of the position of the bar's end precisely one period after the kick, or two periods, or three periods. Those positions will be the same as at the time of the kick and will be completely independent of the kick.